

Problem Statement and Goals

ProgName

Team #, Team Name
Student 1 name
Student 2 name
Student 3 name
Student 4 name

Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

1 Problem Statement

[You should check your problem statement with the [problem statement checklist](#) —SS].

[You can change the section headings, as long as you include the required information. —SS]

1.1 Problem

[What problem are you trying to solve? Why is it important? Do not talk about how you will solve the problem, only what the problem is. AI is rarely part of the problem. The problem is “what” you want to do, not “how” you want to do it. The problem statement should be written in a way that is understandable to a non-technical audience. —SS]

1.2 Inputs and Outputs

[Characterize the problem in terms of “high level” inputs and outputs. Use abstraction so that you can avoid details. —SS]

1.3 Stakeholders

1.4 Environment

[Hardware and Software Environment for the users. The developer environment is summarized as part of the developer plan. —SS]

2 Goals

2.1 Recognition of Post-Treatment Road State

Examine the state of a road after winter maintenance and/or treatment has been done on it, and assign a label to the road state. This is useful for municipalities to assess the quality and completeness of work that is being done with respect to winter maintenance.

2.2 Assessment of Road State Based on Location

Based on the location of the road, assign a label to the road state. Different municipalities have different standards for road maintenance, and the same road state may be considered acceptable in one municipality but not in another. This goal is useful to ensure that results are consistent with pre-existing standards that municipalities have defined.

2.3 Payment Verification Module

Based on the state of the road and the location, determine whether a contractor should be paid for their work. This goal is useful to ensure that the work that contractors do is being assessed in a timely manner.

3 Stretch Goals

- **Contractor Payment Platform:** Build a platform that automatically sends payment to contractors once the AI verifies that a plowing or clearing pass met the required standard. This extends the core payment-verification module from calculating and flagging payments to actually distributing the payments.
- **Dispute and Appeal Workflow:** Provide a structured process for contractors to contest a flagged discrepancy or a rejected pass. Municipal staff can review the supporting imagery and per-segment records to uphold or overturn the decision, with every step logged for auditability.
- **Extended Verification for Sidewalks and Bike Lanes:** Extend the verification pipeline beyond roadways to cover sidewalks and bike lanes, adapting the surface-condition classification and required standards to each type of infrastructure.

- **Road Clearance Duration Estimation:** Estimate how long a road segment will stay clear after treatment, using weather data (such as temperature, precipitation, and wind) together with salting and treatment information, to help municipalities anticipate when re-treatment is needed.

4 Custom Documents (CDs)

[For CAS 741: State whether the project is a research project. This designation, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

[For SE Capstone: List your custom documents (CDs). Potential CDs include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. (The full list is on the course outline and in Lecture 02.) Normally the number of CDs will be two (one per term). Approval of the CDs will be part of the discussion with the instructor for approving the project. The CDs, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?